Ch 2.2: Assessing Model Accuracy

Lecture 3 - CMSE 381

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::
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Weds, Sep 7, 2022

Lec 3

Announcements

Last time:

- Ch 2.1, Vocab day!

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- First homework due TODAY. Covers:
  - Weds 8/31 lecture
  - Fri 9/2 Lecture

Upload a single pdf to D2L

- Office hours
  - Tuesdays, Dr Munch (EGR 1511) 9:30-10:30am
  - Wednesdays, Emily (EGR 1508A) 10:30-noon
  - Thursdays, Dr Munch (EGR 1511) 3:30-4:30pm
  - Fridays, Emily (EGR 1508A) 1-2:30pm

Covered in this lecture

- Mean Squared Error (regression)
- Train vs Test
- Bias Variance Trade off

2.2.1
2.2.2

Quick review of notation

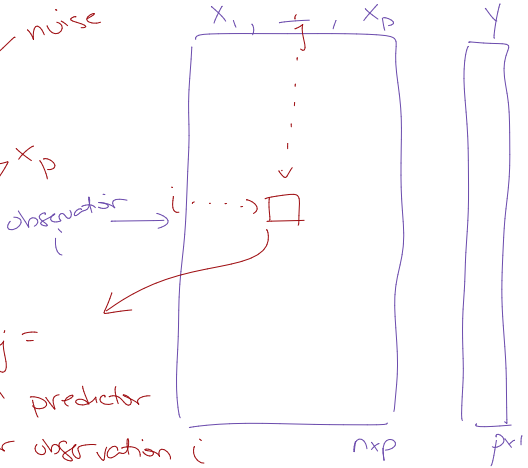
Input data $X_1, \ldots, X_p$

Ground truth $Y = f(\underset{\sim}{X}) + \varepsilon$ ← noise

$\hookrightarrow X_1, \ldots, X_p$

Model $\hat{Y} = \hat{f}(X)$

data pts given $= n$



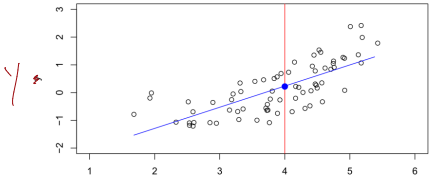
$X_{ij} =$ j^{th} predictor for observation i

Section 1

Mean Squared Error

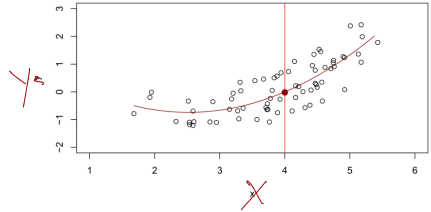
Which is better?

A linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ gives a reasonable fit here



$p = 1$

A quadratic model $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$ fits slightly better.



No free lunch

Goal for today: Way to measure quality of a particular model $\hat{f}$

No single model that is "the best" for all data sets

Want to measure the quality of a model on a given data set

Mean Squared Error

Error in the regression setting

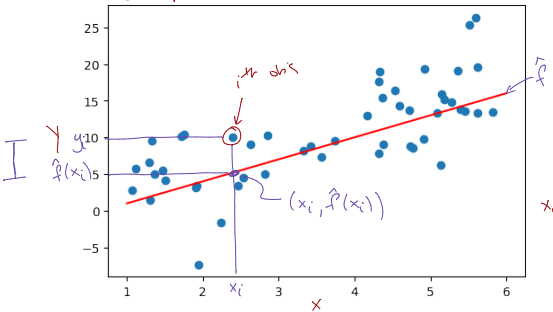
↳ output Y is quantitative

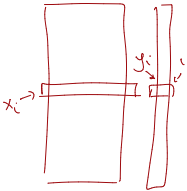
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

data pts

i th obs.

model applied to x values of the i th observation





Group Work

Given the following data, you decide to use the model

$$\hat{f}(X_1, X_2) = 1 - 3X_1 + 2X_2.$$

What is the MSE?

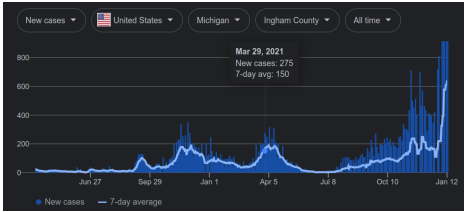
X_1	X_2	Y	$\hat{f}(x_i) = \hat{y}_i$	$\lvert \hat{y}_i - y_i \rvert$	$(\lvert \hat{y}_i - y_i \rvert)^2$
0	7	14	15	1	1
1	-3	-6	-8	2	4
5	2	-10	-10	0	0
-1	1	7	6	1	1

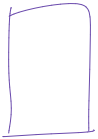
$p = 2$

$n = 4$

$$MSE = \frac{1}{4} 6 = \frac{3}{2}$$

Training MSE





want to know how my model does in terms of predicting data I haven't seen

Train vs test

Training set:
The observations
$\{(x_1, y_1), \cdots, (x_n, y_n)\}$ used to get
the estimate $\hat{f}$

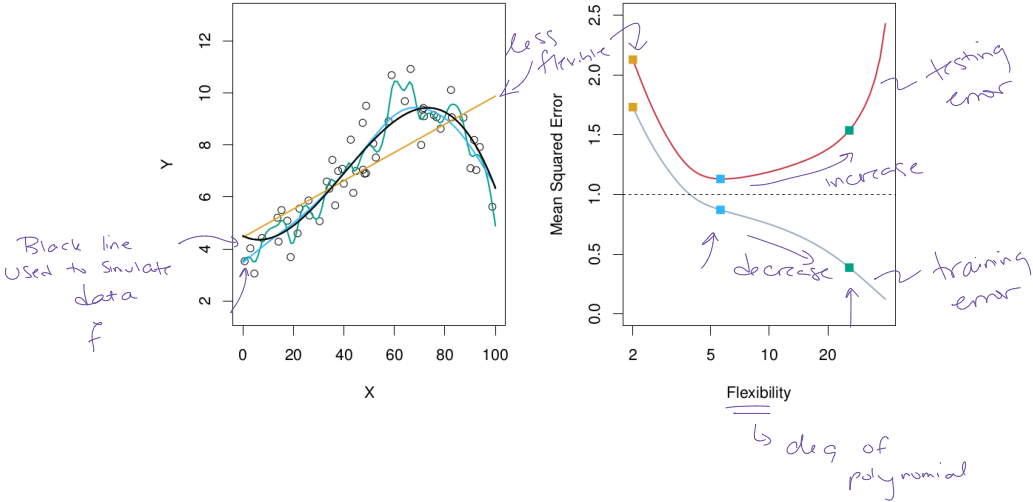
Test set:
The observations
$\{(x'_1, y'_1), \cdots, (x'_{n'}, y'_{n'})\}$ used to
compute the average squared
prediction error

$$\frac{1}{n'} \sum_i (y'_i - \hat{f}(x'_i))^2$$

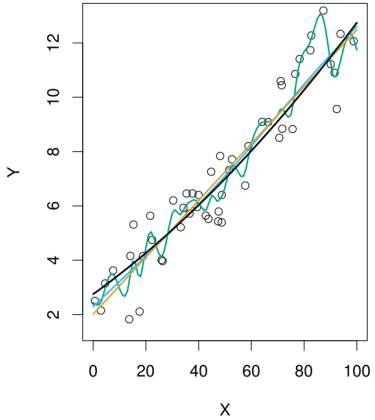
"Data Snooping"
Use test set to train

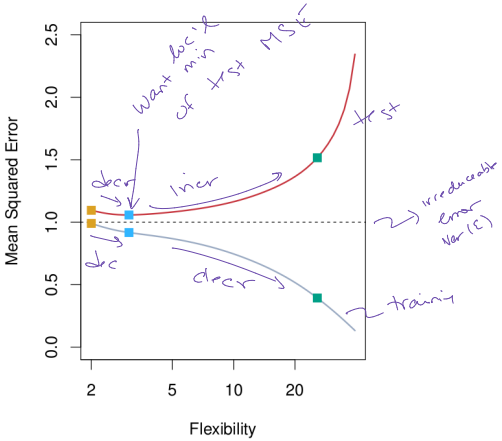
Data not seen while building the model

Why not just get the best model for the training data?

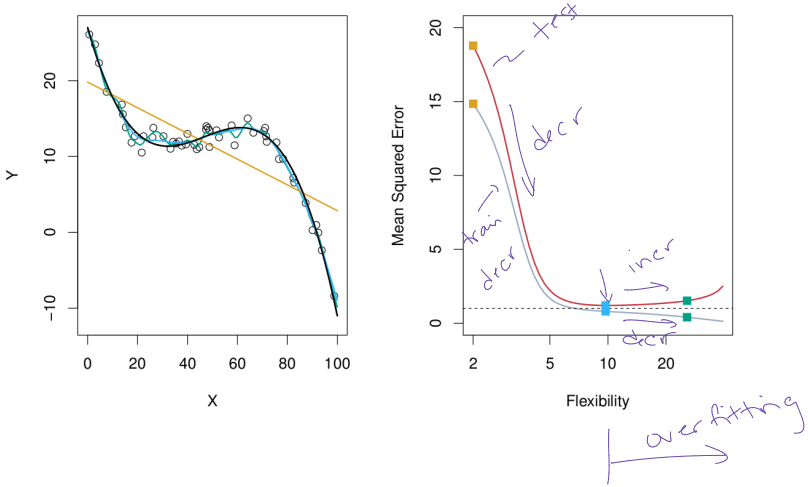


A more linear example



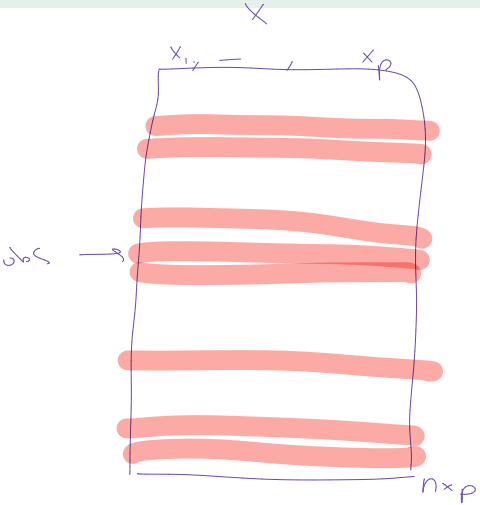


A more non-linear example



A simple solution: Train/test split

More on this in Ch 5



Pick a percentage (30%)

Random subset of 30% of data

Hold it back while training the model

Train model on white rows

Test on red rows

Section 2

Bias-Variance Trade-Off

Bias-variance

$f(x_0) + \varepsilon$ test data fixed model r.v.

$$\underbrace{E(y_0 - \hat{f}(x_0))^2}_{\text{True test MSE}} = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{Variance of model}} + \underbrace{\left[\mathrm{Bias}(\hat{f}(x_0))\right]^2}_{\text{Bias of model}} + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{Irreducible error}}$$

every value is positive

↳ True MSE $\geq$ Irreducible error

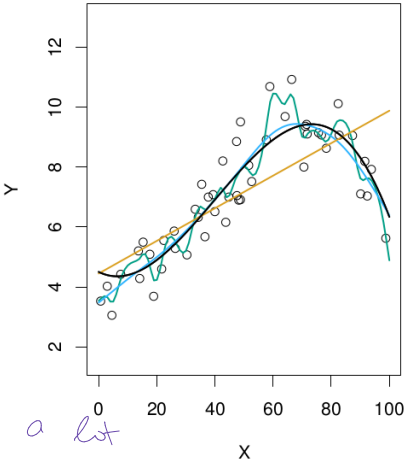
Variance

$Var(\hat{f}(x_0))$

High variance:

Small changes in the training data result in large changes in $\hat{f}$

Variance: the amount by which $\hat{f}$ would change if we estimated it using a different training data set.



Green squiggly version

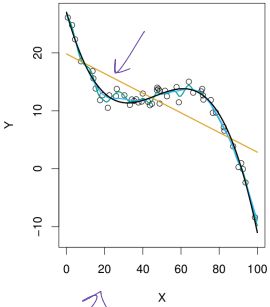
High variance b/c more/diff data pts change the wiggles a lot

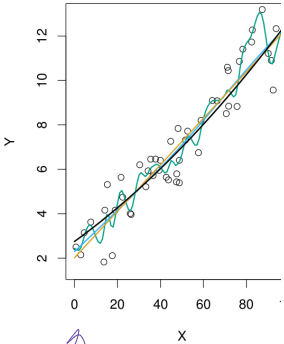
Linear model low variance

Bias

Bias: the error that is introduced by approximating a (complicated) real-life problem by a much simpler model.

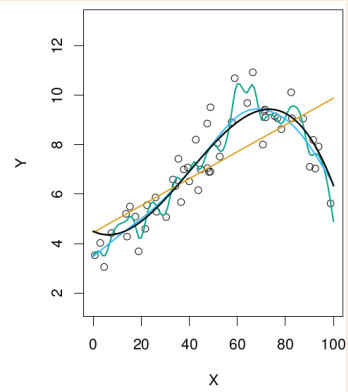
Linear regression: too simple for this data
$\Rightarrow$ high bias

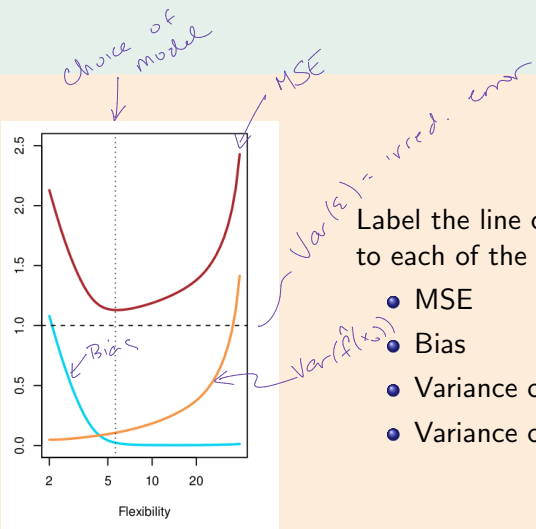




Lin. reg has low bias b/c it is a reasonable fit for data

Group work



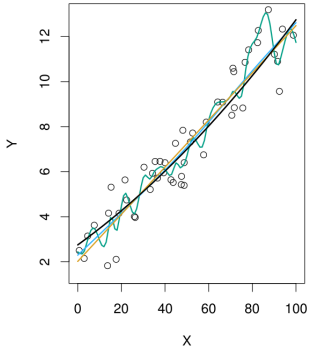


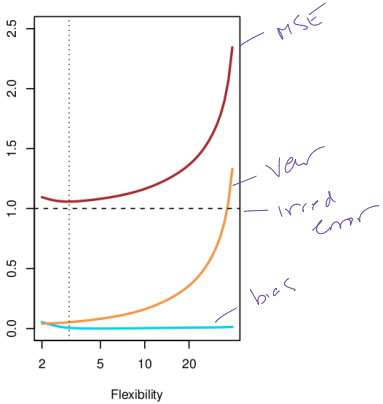
Label the line corresponding to each of the following:

- MSE
- Bias
- Variance of $\hat{f}(x_0)$
- Variance of ε

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

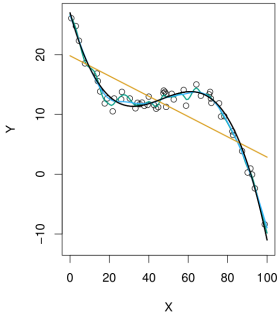
Another example

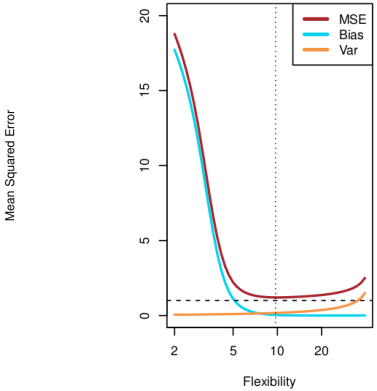




$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

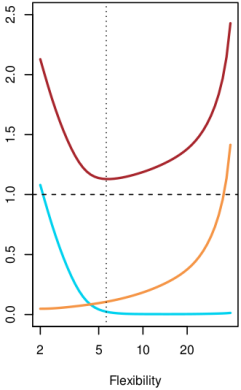
Yet another example





$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Bias-variance trade off



Bias decr

Var increasing

Relative speed ⇝ MSE

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Group work: coding

See jupyter notebook

Next time

- Wednesday:
  - Homework due midnight on D2L
- Friday:
  - 3.1 Linear Regression